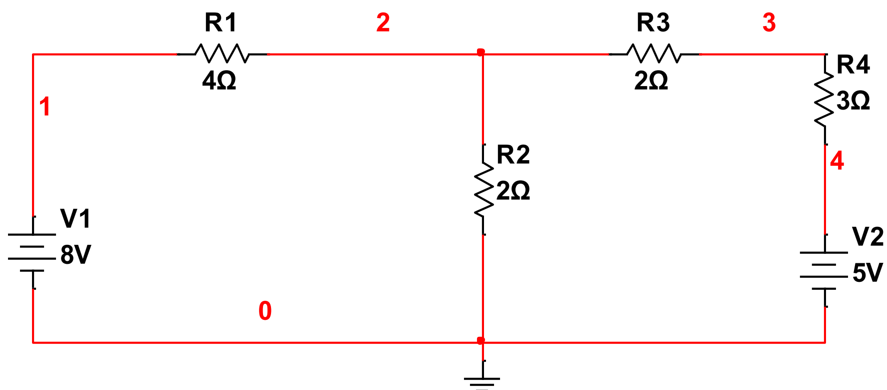
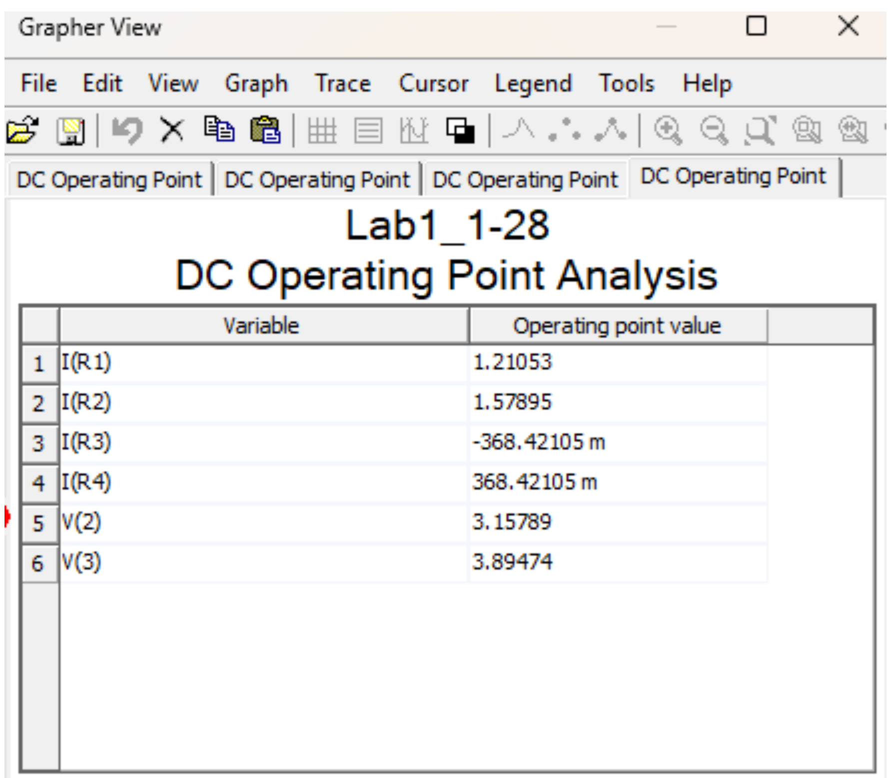
ET 252/L Lab 1

Perla Dominguez

Introduction: Simulating a two loop DC circuit to confirm handwritten circuit analysis results.



Results: Simulation for DC operating point



Lab1_1-28
DC Operating Point Analysis

	Variable	Operating point value
1	I(R1)	1.21053
2	I(R2)	1.57895
3	I(R3)	-368.42105 m
4	I(R4)	368.42105 m
5	V(2)	3.15789
6	V(3)	3.89474

ET 252/ Lab 2

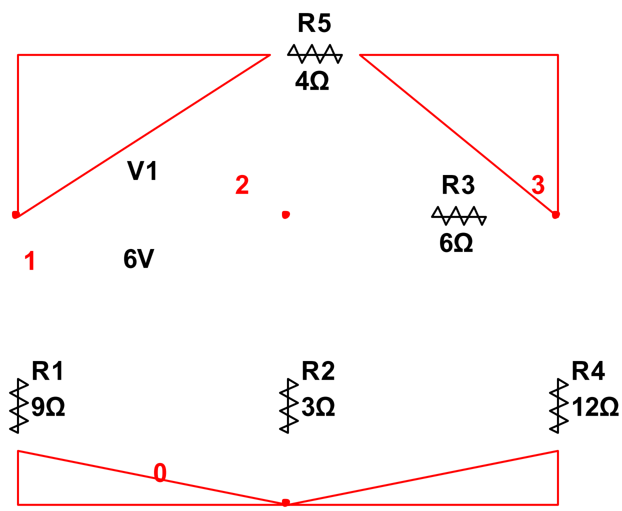
Perla Dominguez

2/2/2026

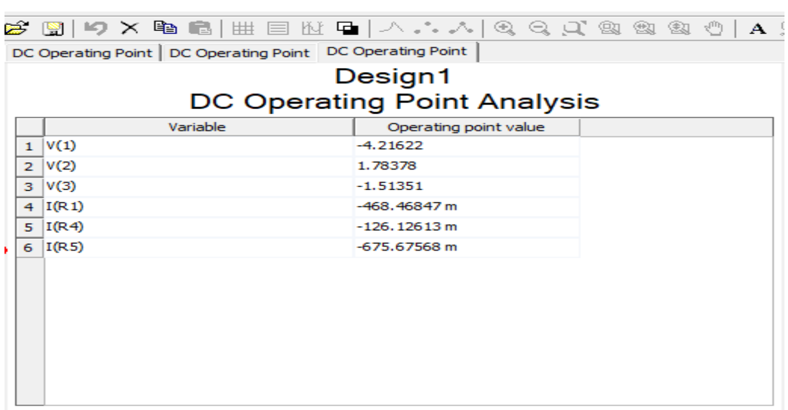
[redacted]

Instruction: Modeled Nodal analysis on Multisim, as well as confirming voltage values in MatLab. Some values appear negative due to Multisim environment.

Nodal Analysis Schematic



Simulation Results From Table



DC Operating Point | DC Operating Point | DC Operating Point

Design1

DC Operating Point Analysis

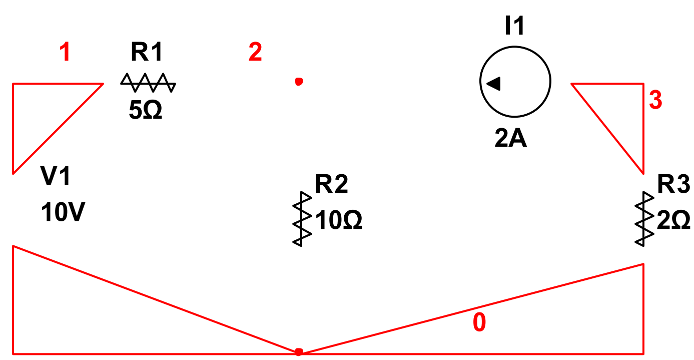
	Variable	Operating point value
1	V(1)	-4.21622
2	V(2)	1.78378
3	V(3)	-1.51351
4	I(R1)	-468.46847 m
5	I(R4)	-126.12613 m
6	I(R5)	-675.67568 m

ET 252 LAB 3

Perla Dominguez, [redacted]

Objective: We modeled a two mesh schematic on Multisim to verify voltage and current values through the nodes and branches. In this example, we wrote KCLs on paper to solve for V2. V1 was given, and V3 was hand calculated as well. This Multisim simulation proves that our math was correct on paper.

Schematic:



DC Operating Port Analysis Table:

Design1

DC Operating Point Analysis

	Variable	Operating point value
1	V(1)	10.00000
2	V(2)	13.33333
3	V(3)	-4.00000
4	I(R1)	-666.66667 m
5	I(R2)	1.33333
6	I(R3)	-2.00000

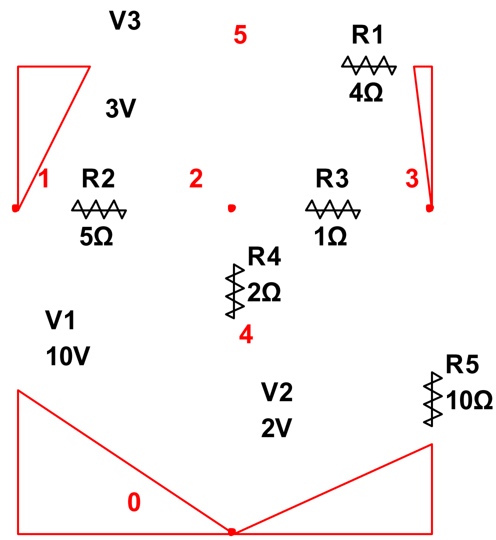
ET 252: Lab 4

Perla Dominguez [redacted]

2/4/2026

Instruction: Using Multisim to simulate three mesh circuit and to find voltage across each node, labeled as V1, V2, V3, in this example. In addition, we also test for current across each node modeling a virtual simulation of handwritten calculations.

Three Mesh Schematic Simulation:



Resistor Values R1 – R5:

DC Operating Point | DC Operating Point

Design1
DC Operating Point Analysis

	Variable	Operating point value
1	I(R1)	1.60425
2	I(R2)	872.58687 m
3	I(R3)	-945.94595 m
4	I(R4)	1.81853
5	I(R5)	658.30116 m

Selected Diagram:DC Operating Point Analysis

Voltage Across Nodes Simulation:

File Edit View Graph Trace Cursor Legend Tools Help

DC Operating Point | DC Operating Point | DC Operating Point | DC Operating Point | DC Operating Point

Design1
DC Operating Point Analysis

	Variable	Operating point value
1	V(1)	10.00000
2	V(2)	5.63707
3	V(3)	6.58301

Selected Diagram:DC Operating Point Analysis

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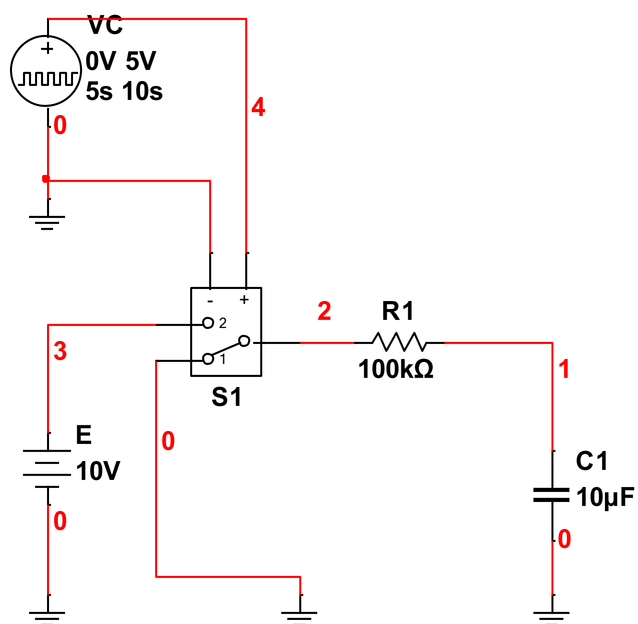
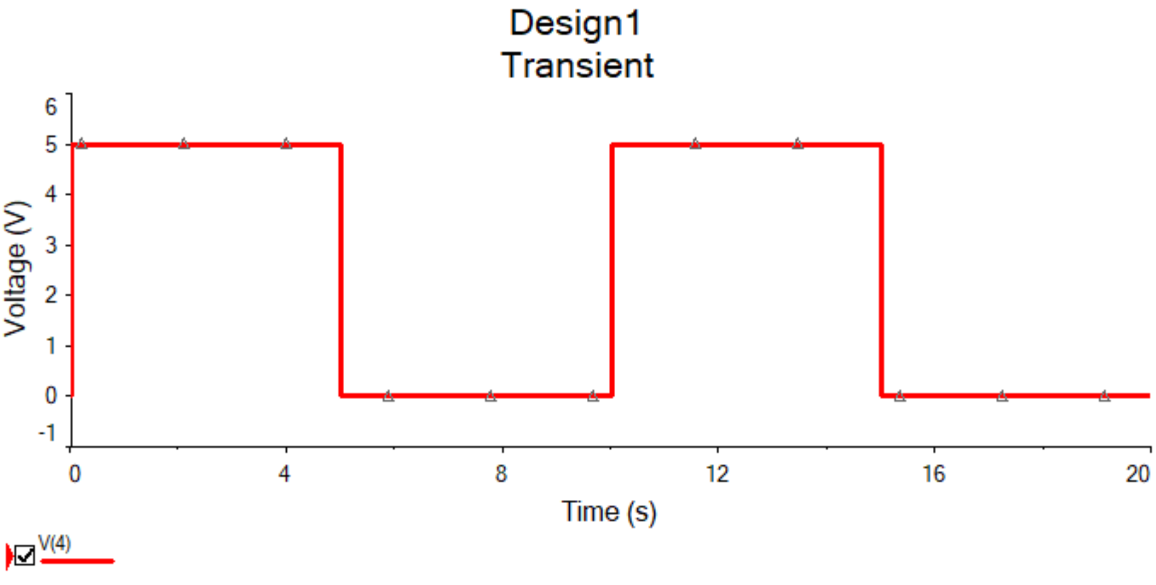
Lab 5

Objective: Using mutism, we modeled three different simulations to demonstrate a capacitor charging and displaying a waveform in seconds. We used three different kinds of transient graphs.

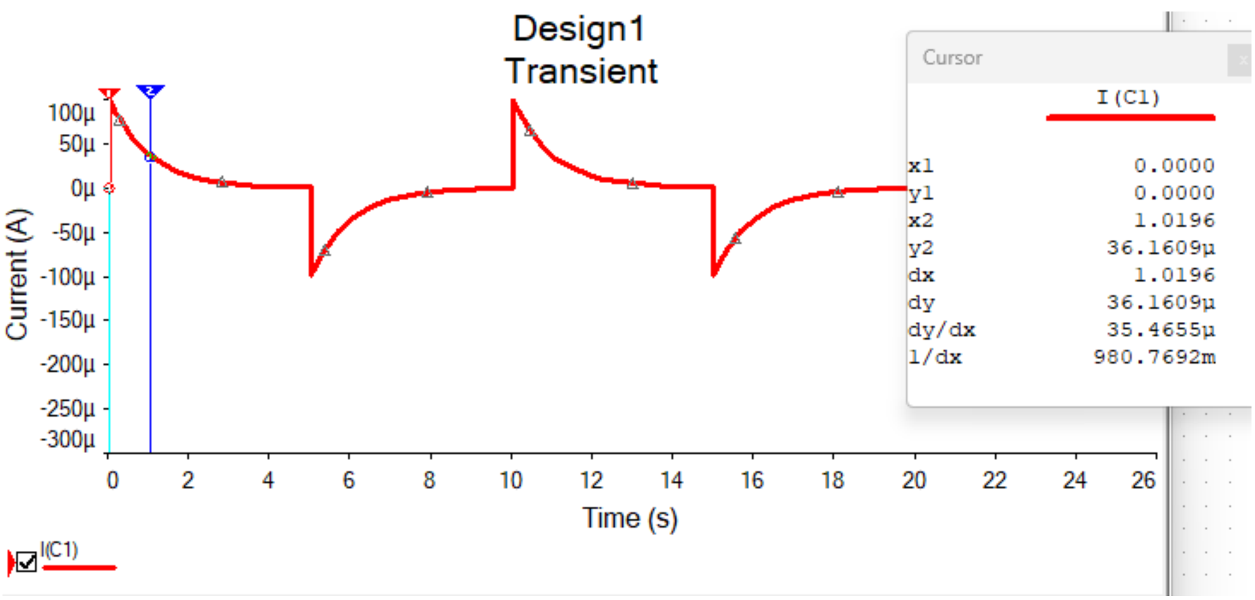
Schematic:



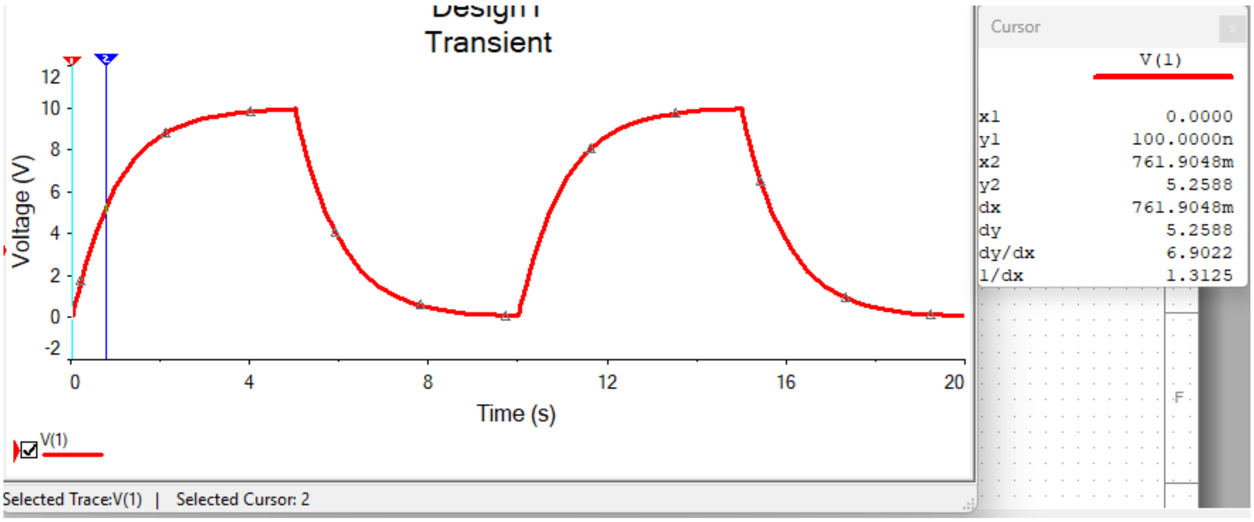
Square Waveform at Node 4:



Transient Simulation: Capacitor Charging and Discharging Current



Transient Simulation: Capacitor Voltage



LAB EXPIREMENT 6

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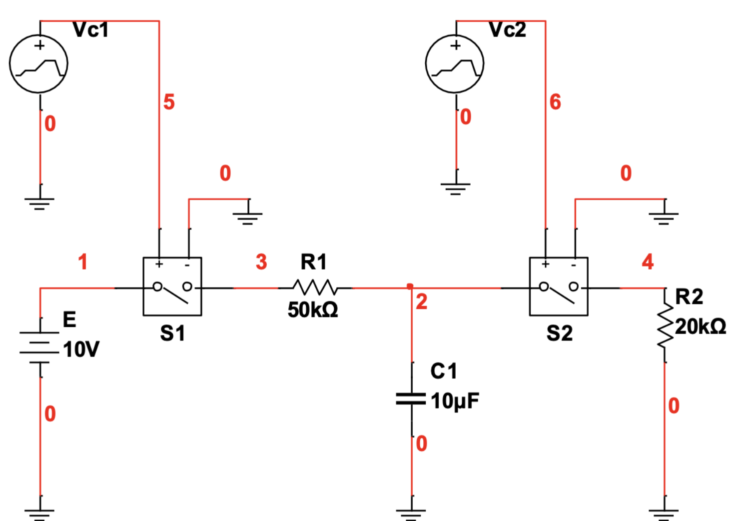
ET 252 : CIRCUITS II

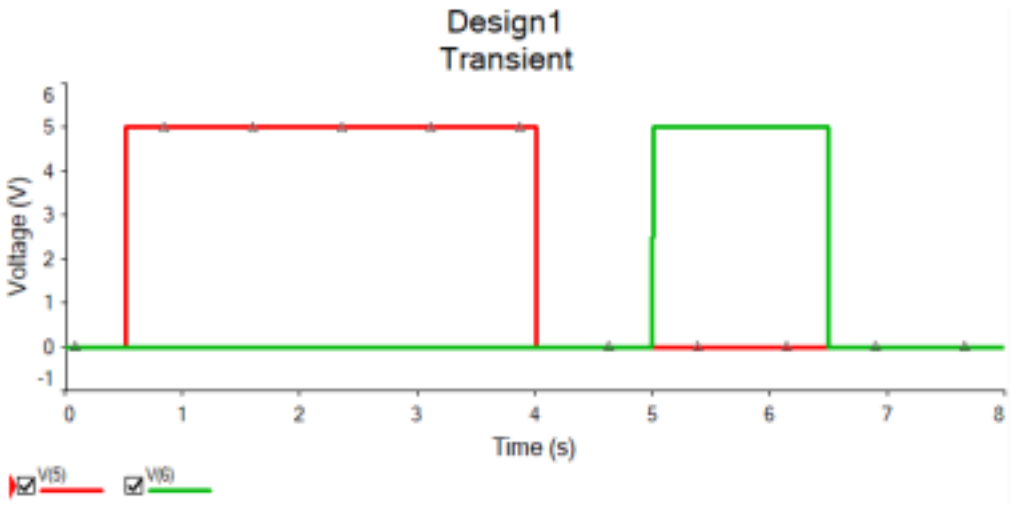
3/4/2026

Objective: In this lab, we focused on the transient response of an RC circuit, specifically observing the charging and discharging phases of a 10 microfarad capacitor. The circuit was powered by a 10 volt DC source and utilized two switches, S1 and S2, to control the flow of current through a 50 kilohm charging resistor and a 20 kilohm discharging resistor. As shown in the simulation results, closing S1 at 0.5 seconds initiated a charging phase where the capacitor voltage rose toward 10 volts , characterized by an initial current spike of approximately 100 microamps. Conversely, closing S2 at 5.0 seconds triggered a rapid discharge through the second resistor, resulting in a sharp negative current spike of approximately negative 500 microamps as the stored energy was released to ground

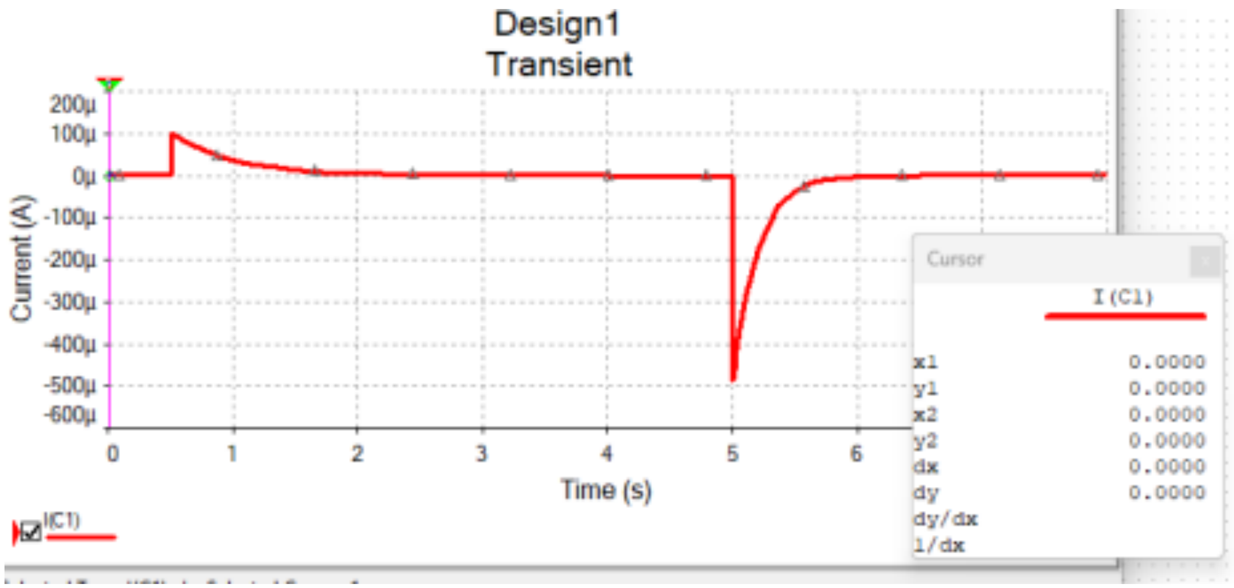
Schematic:



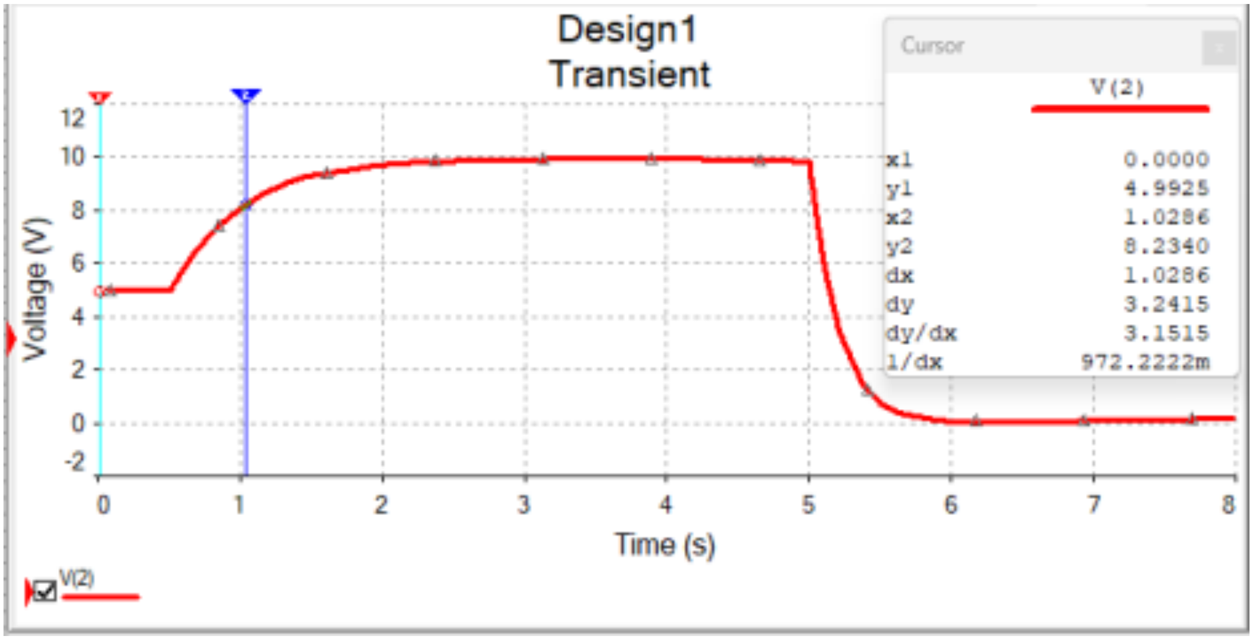
Transient Simulation 1:



Transient Simulation 2:



Transient Simulation 3:


Design1
Transient
Cursor
V(2)
x1 0.0000
y1 4.9925
x2 1.0286
y2 8.2340
dx 1.0286
dy 3.2415
dy/dx 3.1515
1/dx 972.2222m
Voltage (V)
Time (s)
V(2)

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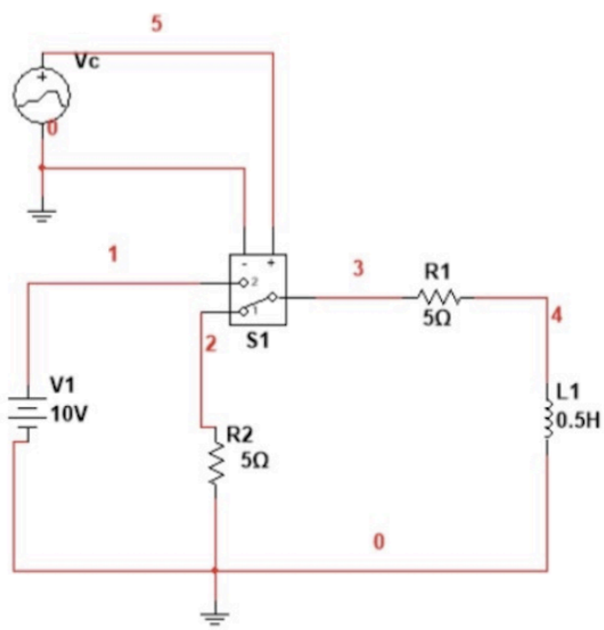
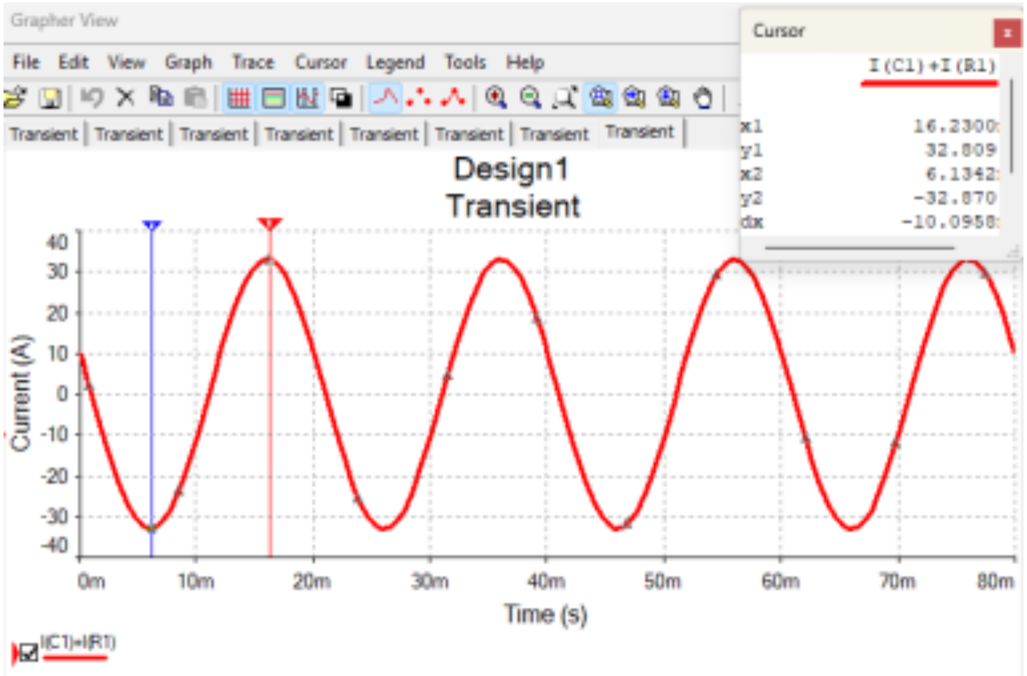
Lab Experiment 7

Objective: In this lab experiment, we used a signal voltage source to model a parallel circuit to simulate steady-state behavior using sinusoidal excitation. The goal is to analyze the relationship between voltage and the combined current of resistive and capacitive branches in an AC environment

Schematic: In this lab experiment, we have modeled an RL circuit. We modeled using a transient simulation to verify the storage and decay time constraints. We also verified voltage and current for our indicator.





Grapher View
File Edit View Graph Trace Cursor Legend Tools Help
Transient Transient Transient Transient Transient Transient Transient Transient
Design1
Transient
Current (A)
Time (s)
I(C1)+I(R1)
Cursor
I(C1)+I(R1)
x1 16.2300
y1 32.809
x2 6.1342
y2 -32.870
dx -10.0958

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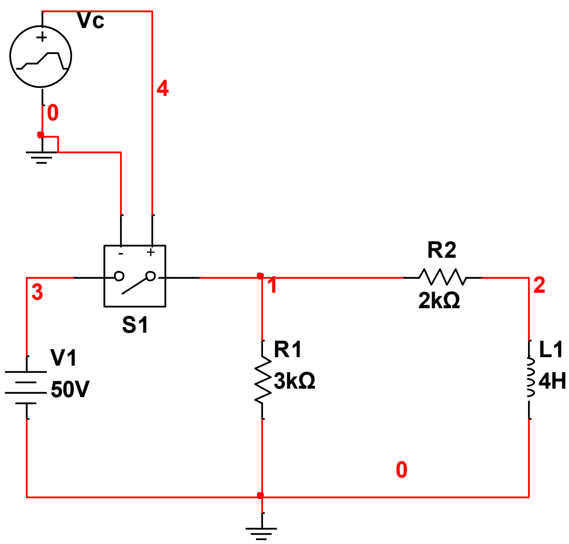
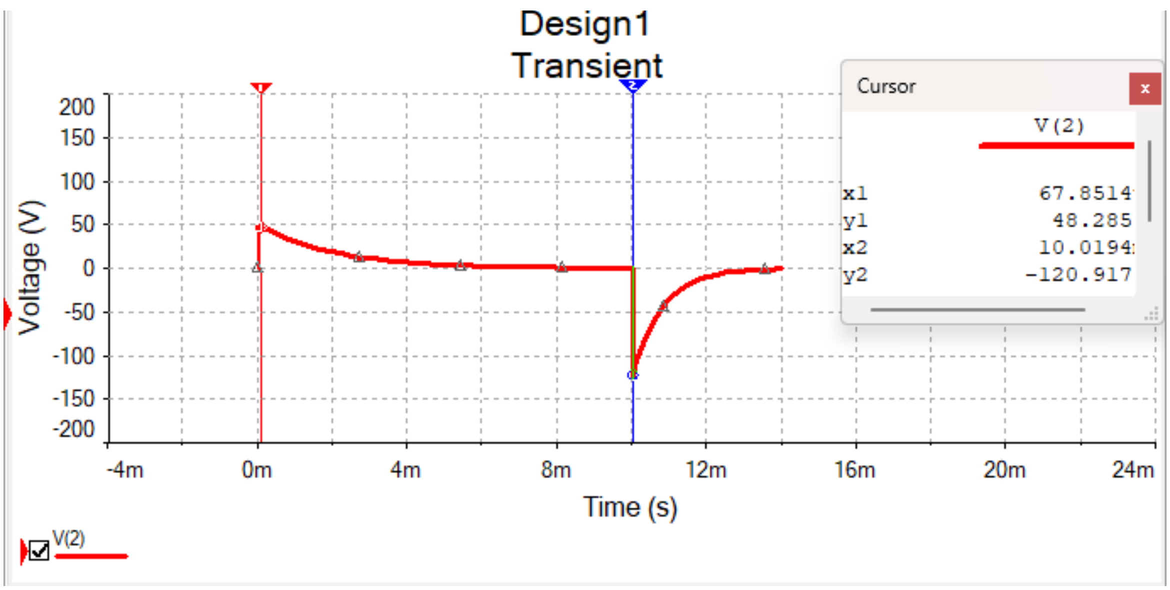
Lab 8

Objective: In this lab, we proved the in class example shown in the circuit below by modeling it on Multisim. We captured four simulations, two of them being the inductor behavior and the other two capturing the resistor behavior. Overall this lab was informative in visualizing the behavior at these scenarios.

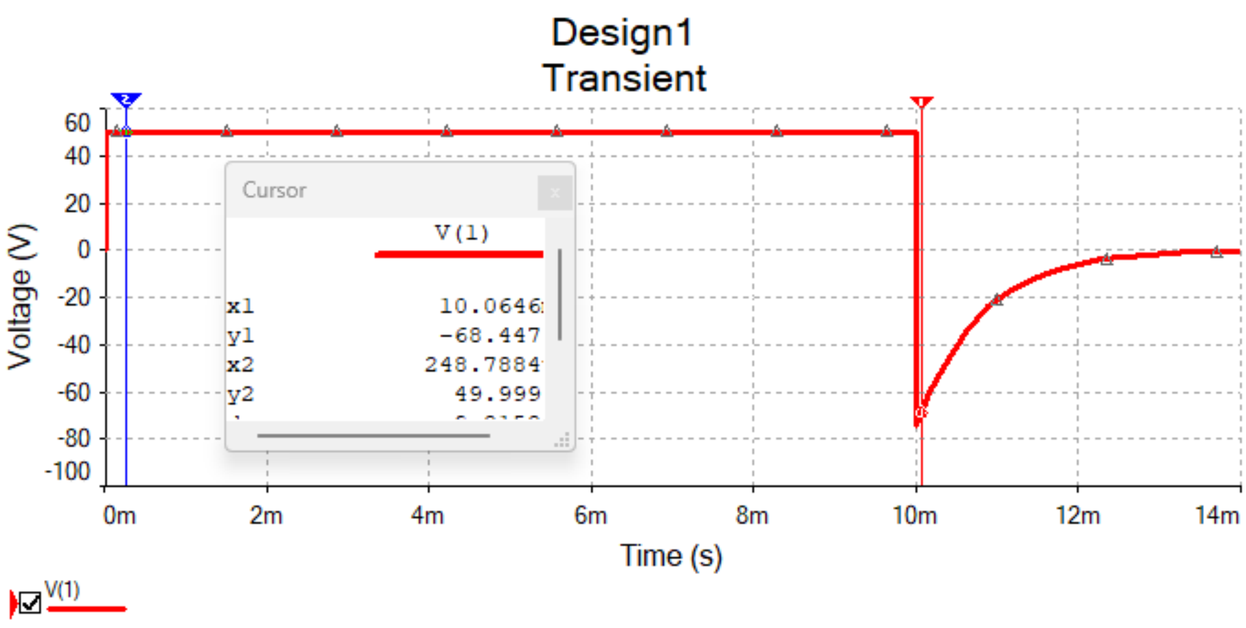
Schematic:



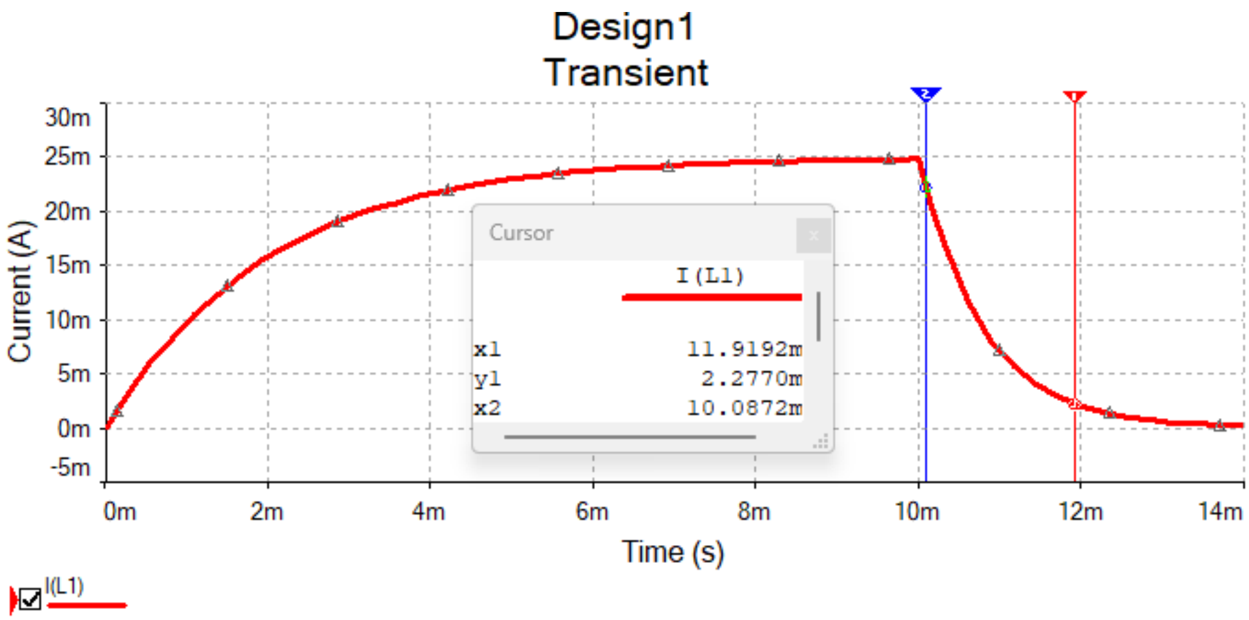
Simulation 1: VR2



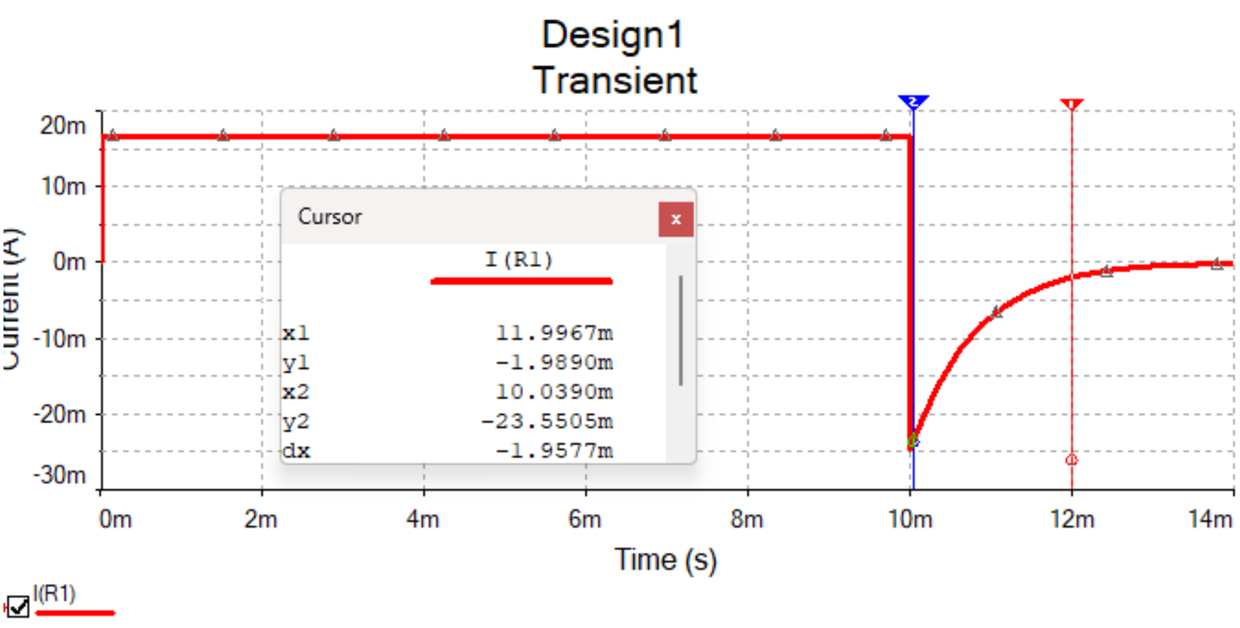
Simulation 2: VR1



Simulation 3: Inductor Current



Simulation 4: Inductor Resistance



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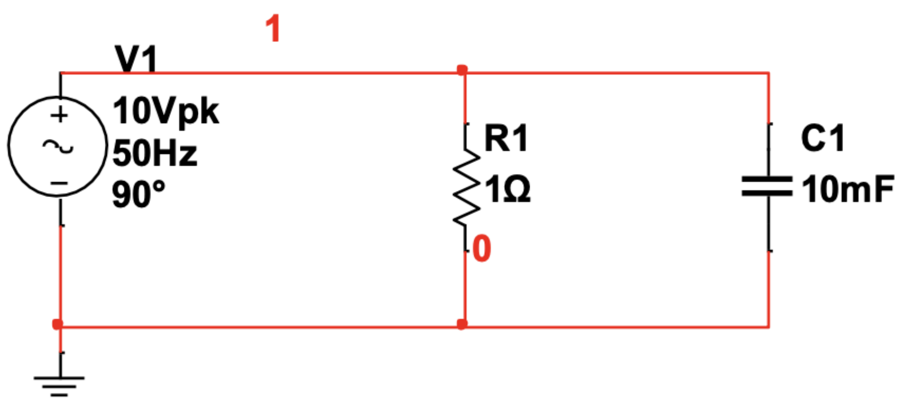
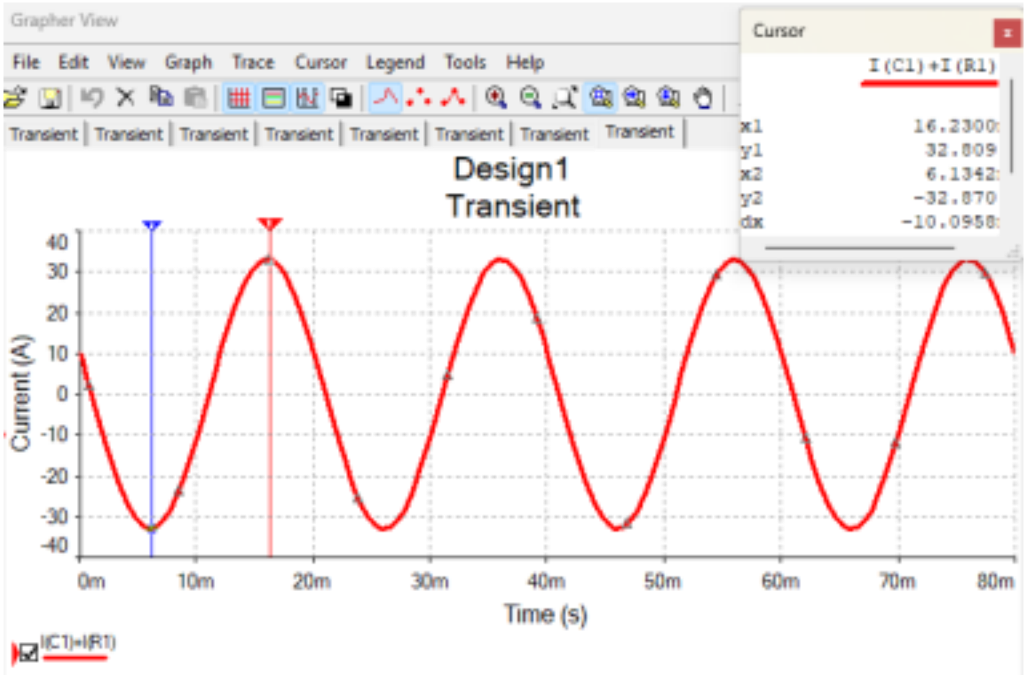
4/15/2026

Lab Experiment 9

Objective: In this lab experiment, we used a signal voltage source to model a parallel circuit to simulate steady-state behavior using sinusoidal excitation. The goal is to analyze the relationship between voltage and the combined current of resistive and capacitive branches in an AC environment

Schematic:




Grapher View
File Edit View Graph Trace Cursor Legend Tools Help
Transient Transient Transient Transient Transient Transient Transient Transient
Design1
Transient
Cursor
I(C1)+I(R1)
x1 16.2300
y1 32.809
x2 6.1342
y2 -32.870
dx -10.0958
Current (A)
Time (s)
0m 10m 20m 30m 40m 50m 60m 70m 80m
40 30 20 10 0 -10 -20 -30 -40
I(C1)+I(R1)

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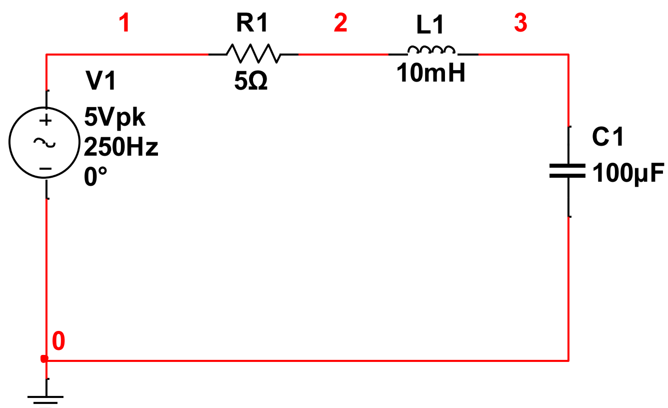
LAB 10

Objective: We analyzed an RLC circuit and verified calculations on multisim by using transient simulations.

Schematic:



:

1. Compute admittance of RC parallel combination
   xL = 2pi(250)(10 x 10 ^-3) = 15.71 ohm
   xC = 1/ 2pi(250)(100 x 10 ^-6) = 6.37 ohm
   x1 = 15.71 – 6.37 = 9.34

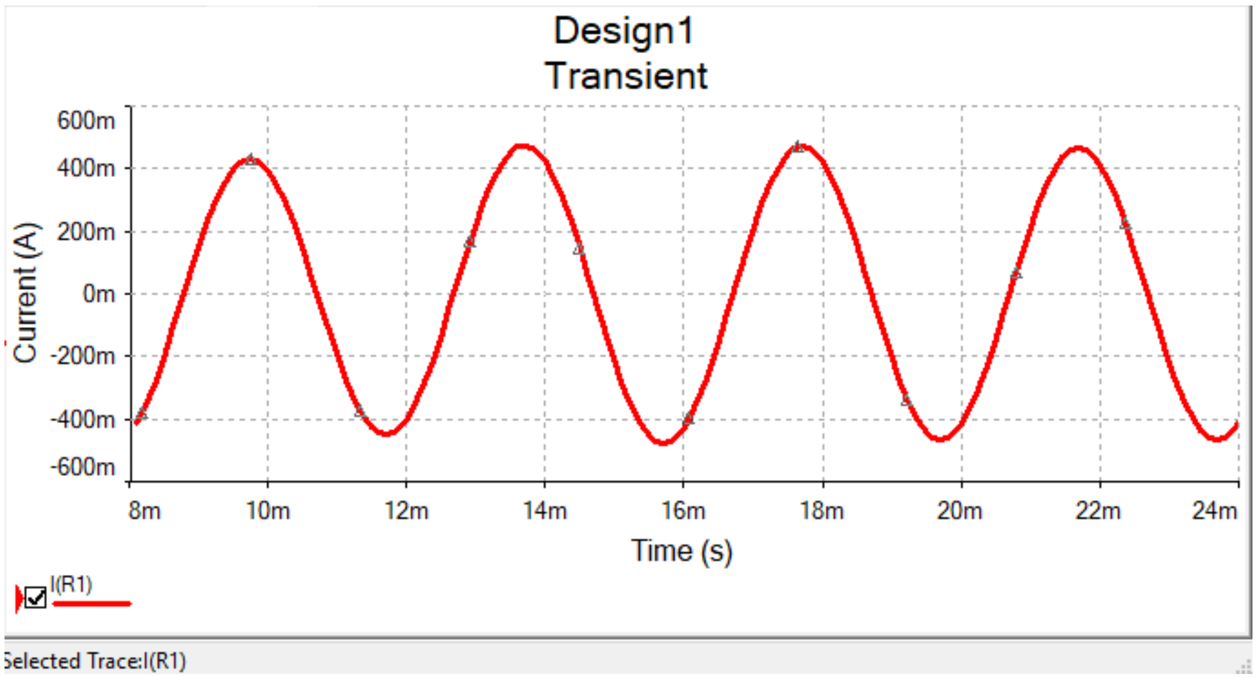
2. Magnitude and Phase Angle
   Magnitude: sqrt (5^2 + 9.34^2) = 10.60 ohm
   Phase Angle: tan-1(9.34/5) = 61.8 deg

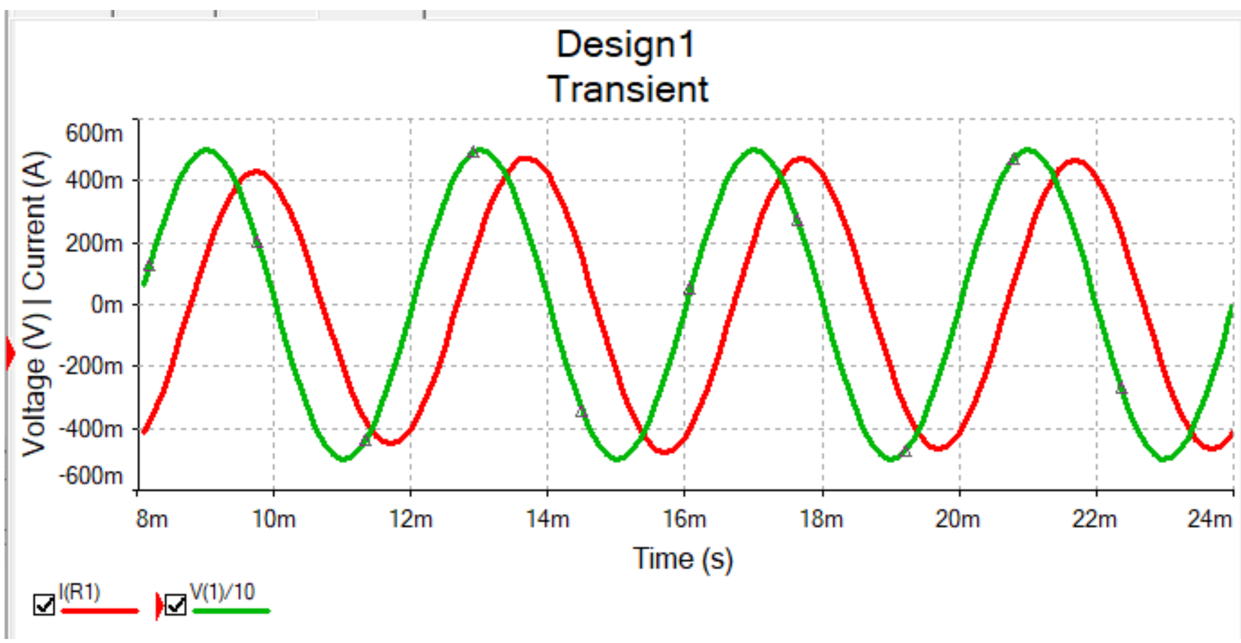
3. Measured Peak

   Ip = 5/10.60 = 0.47 A

4. (725.21 * 10^-6/ 0.004) * 360 deg = 65.27 deg

Transient Simulations:





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5/4/2026

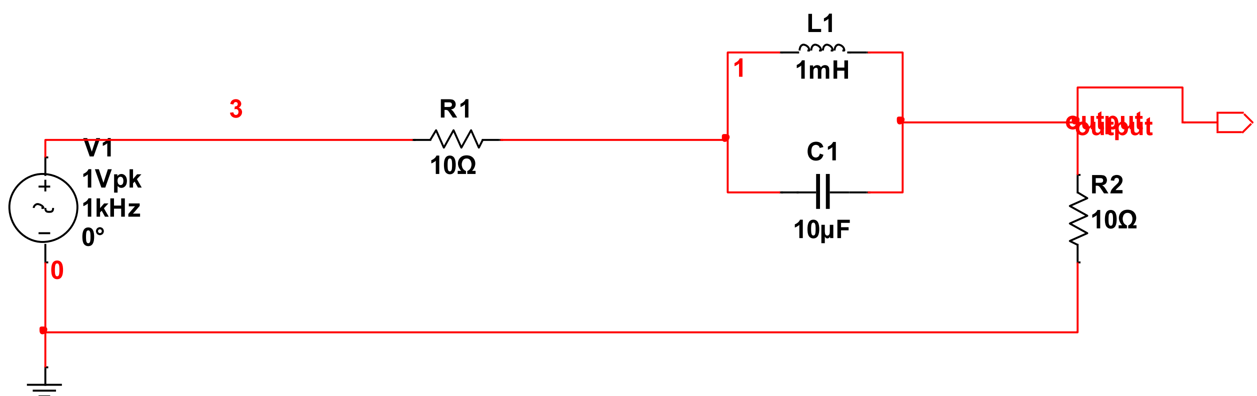
ET 252/L

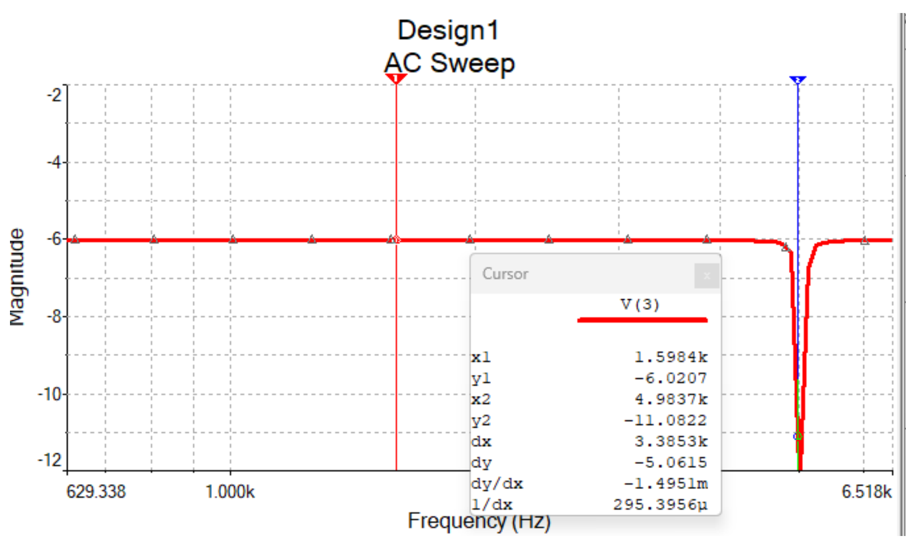
Lab 11

Objective: This lab was broken down into two parts. On PART 1, we set up a notch filter on Multisim. This filter eliminates any frequency that is not desirable. This was achieved using an AC Sweep Simulation. We also calculated resonance frequency of parallel inductor and capacitor by hand and checked it with the software. In PART 2, we modeled and RLC Series low pass filter with a range. This was modeled using a sweep type Decade, 100 points per decade, and vertical scale decibel.

Part 1: Filter 1

Schematic:

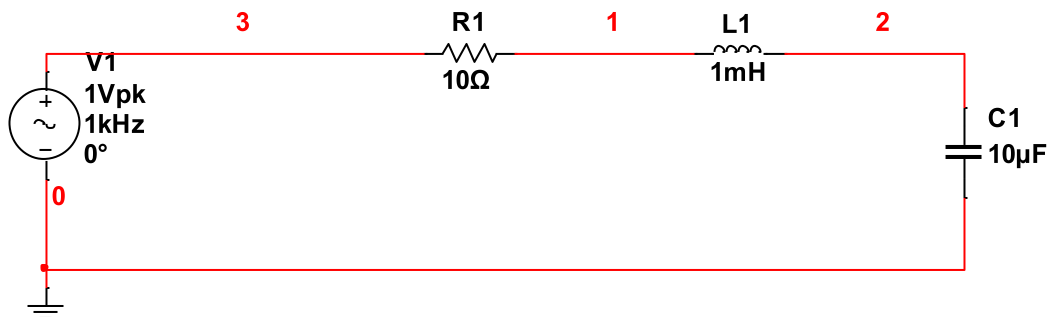


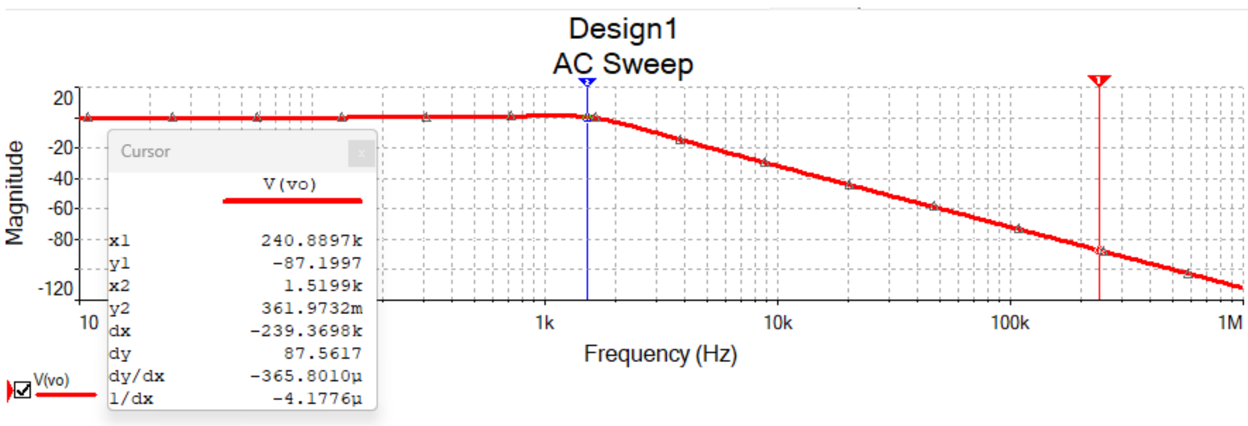


Resonant Frequency: f(0) = 1.59kHz

Part 2: Filter 2

Schematic:





X1 = 240.99